

Forecasting cryptocurrency price Trends using ARIMA and LSTM

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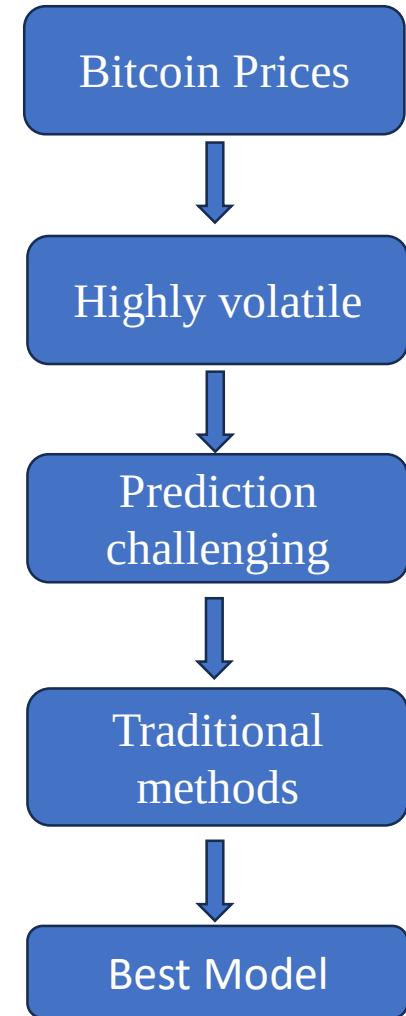
Introduction

- Cryptocurrency prices will fluctuate based on market demand.
- The project focus on predicting bitcoin prices.
- I used two different approach's ARIMA a statistical model which is used to capture trends and another one is LSTM deep learning model to learn complex patterns



Problem Statement

- Bitcoin prices are highly volatile and prices can change rapidly over the time which makes the price prediction challenging
- The traditional methods are struggle to capture the sudden changes.
- There is a need to evaluate the models which can handle both trend and fluctuations
- In this Project I will compare both models to identify the better performing model for prediction



Data Collection and Cleaning

- The Bitcoin Data was collected using Binance API
- The collected was in BTC/USD which has interval of 1-minute and duration of 60 days around 80000 records
- The dataset has features like open, High, Low, Close , volume, open time, close time, Quote asset volume, Number of trades"
- UNIX timestamp conversation of data to date and time format
- Converted the variables like OHLC and volume to numeric format and removed the unnecessary columns
- Checked for duplicate and missing values in data and prepared data for both ARIMA and LSTM models

Exploratory Data Analysis-Trend and Stationarity Insights



- In the Graph the bitcoin close prices are fluctuating rapidly
- There is no constant mean the values are shifting across ranges



- From the rolling mean and standard deviation graph both mean and standard deviation are changing continuously.
- Which indicates the data is non-stationary

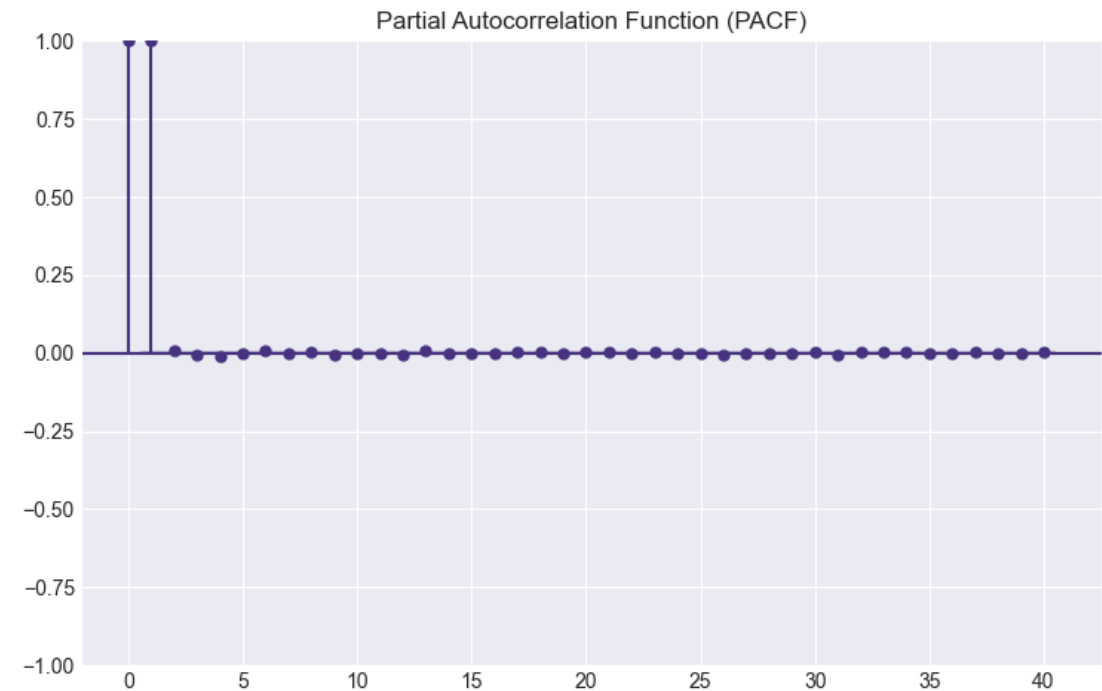
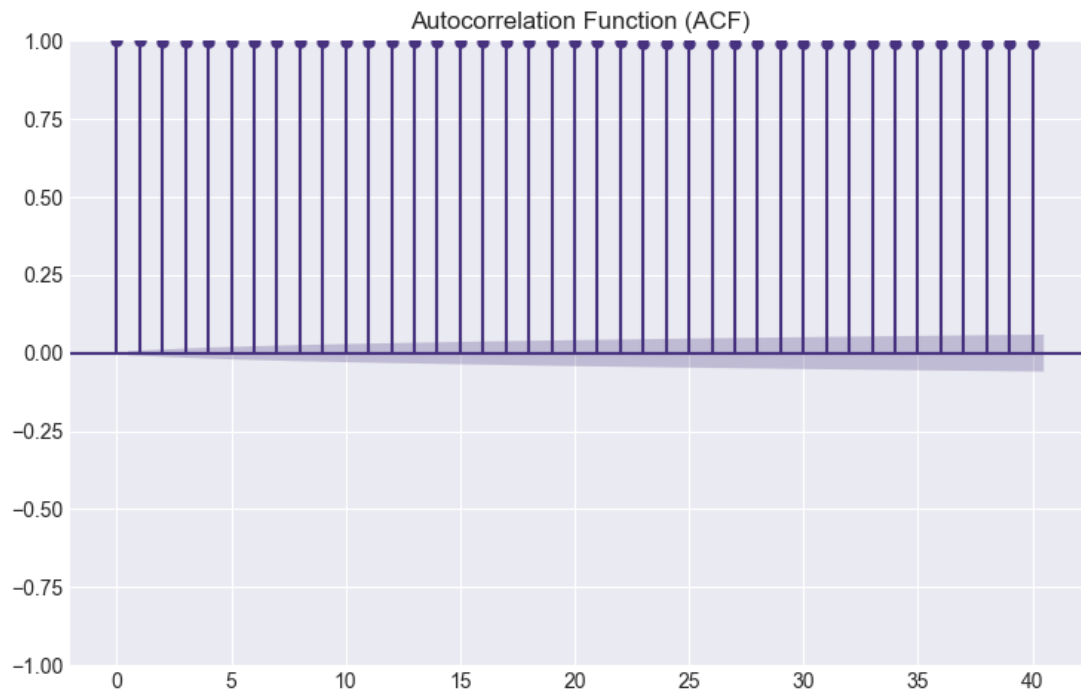
Stationarity Test

- Initially the EDA shows bitcoin data is not stationary due to price fluctuations.
- To test the data is stationary or not , applied Augmented Dickey-Fuller (ADF) test
- The original close price series showed a p-value of 0.0008, which is less than 0.05, indicating that the data is stationary.
- Further after Applying the differencing resulting in lower p-value 0.00 confirms the data is strong stationary suitable for time series modelling.

```
--- ADF Test for Close Price ---  
ADF Statistic: -4.1582  
p-value: 0.0008  
Critical Values:  
    1%: -3.4304  
    5%: -2.8616  
   10%: -2.5668
```

```
--- ADF Test for Differenced Close Price ---  
ADF Statistic: -38.0847  
p-value: 0.0000  
Critical Values:  
    1%: -3.4304  
    5%: -2.8616  
   10%: -2.5668
```

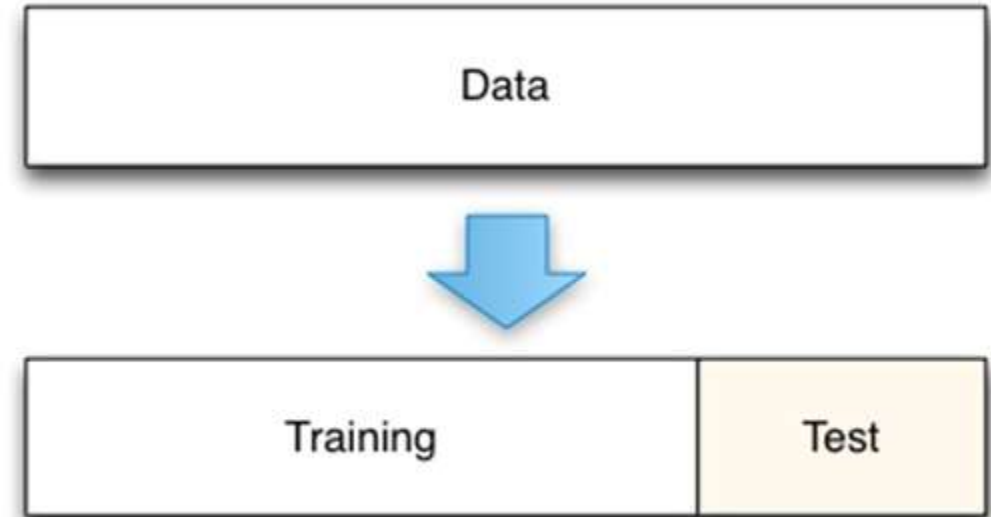
ACF & PACF Analysis for ARIMA parameter Selection



- The ACF plot shows Some strong correlation across many lags
- Which indicates the past values highly influence on current values
- From this graph is shows there is no sharp cut off so the moving average component is set to $q = 0$.
- The PACF which shows a sharp spike at lag 1 then drops to zero
- Since previous value directly influence the current value so $p=1$
- Applied differencing to make the data stationary is $d=1$
- Initial ARIMA model is set to $(p,d,q)-(1,1,0)$

ARIMA Model Development

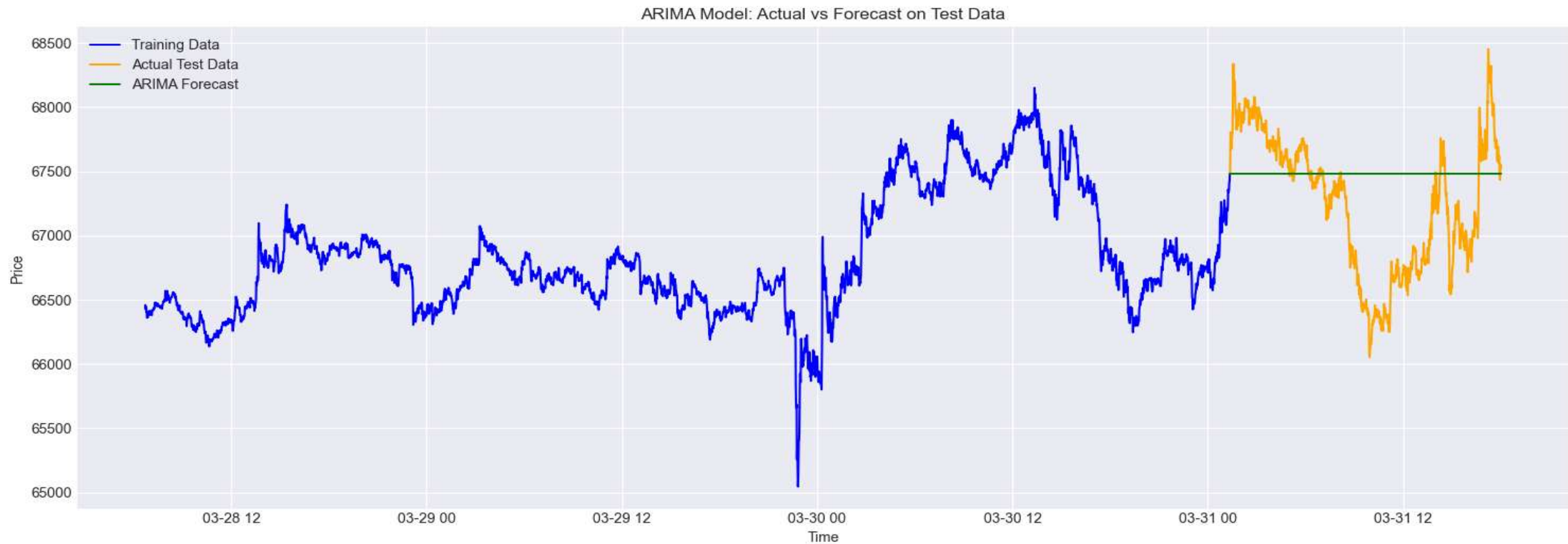
- Then the dataset was split into training(69081 ,80.0%) and testing (17271 ,20.0%)
- Initial ARIMA parameters are identified based on ACF and PACF ARIMA(1,1,0)
- Hyperparameter Tuning performed different p,d,q combinations
- Final ARIMA(0,1,0) model selected based on lower RMSE
- Predictions were generated on test data



Model Performance

- Evaluated using MAE, RMSE and MAPE
- MAE and RMSE may appear large but they are small when compared to overall bitcoin prices and low MAPE will indicate high prediction accuracy

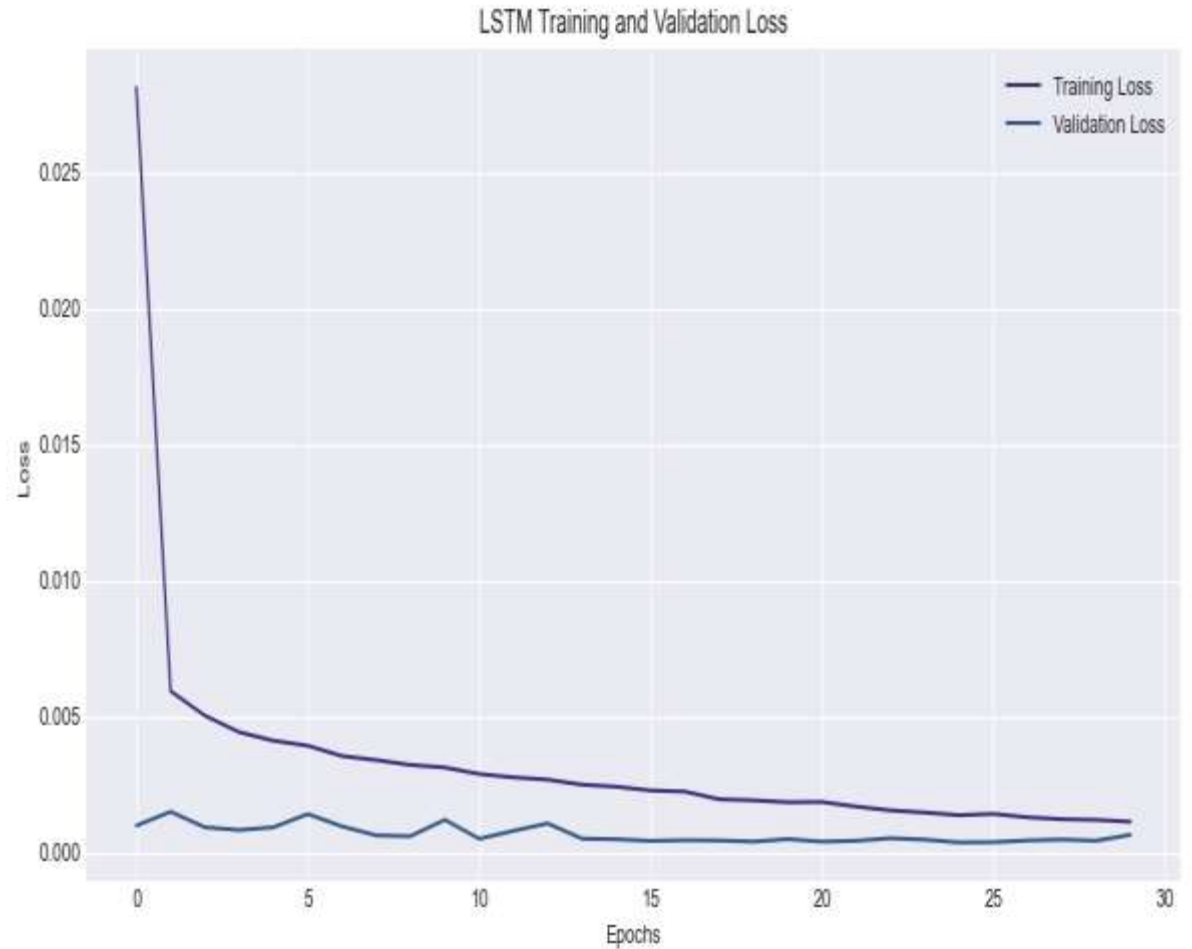
Model Performance	
MAE	466.92
RMSE	577.49
MAPE	0.70



- Arima graph showing a flat line because during model Tuning the best model selected as ARIMA(0,1,0) which indicates model relies on differencing $d=1$ and no moving component(MA) and AutoRegressive component (AR)
- The ARIMA model performs well in capturing the overall price level, but it does not accurately capture short-term fluctuations due to the high volatility of cryptocurrency markets

LSTM Model Development

- Here I used 2 LSTM layers of 50 units each with dropout 0.2 which reduce overfitting
- The model was trained on Adam optimizer and MSE loss function with 32 batch size , 50 epochs and 10% split validation
- In the graph both training and validation loss decreases consistently, which shows the model is learning patterns effectively.
- Also, both curves are closer to each other which indicating good generalization and no overfitting which shows the model is well trained for prediction

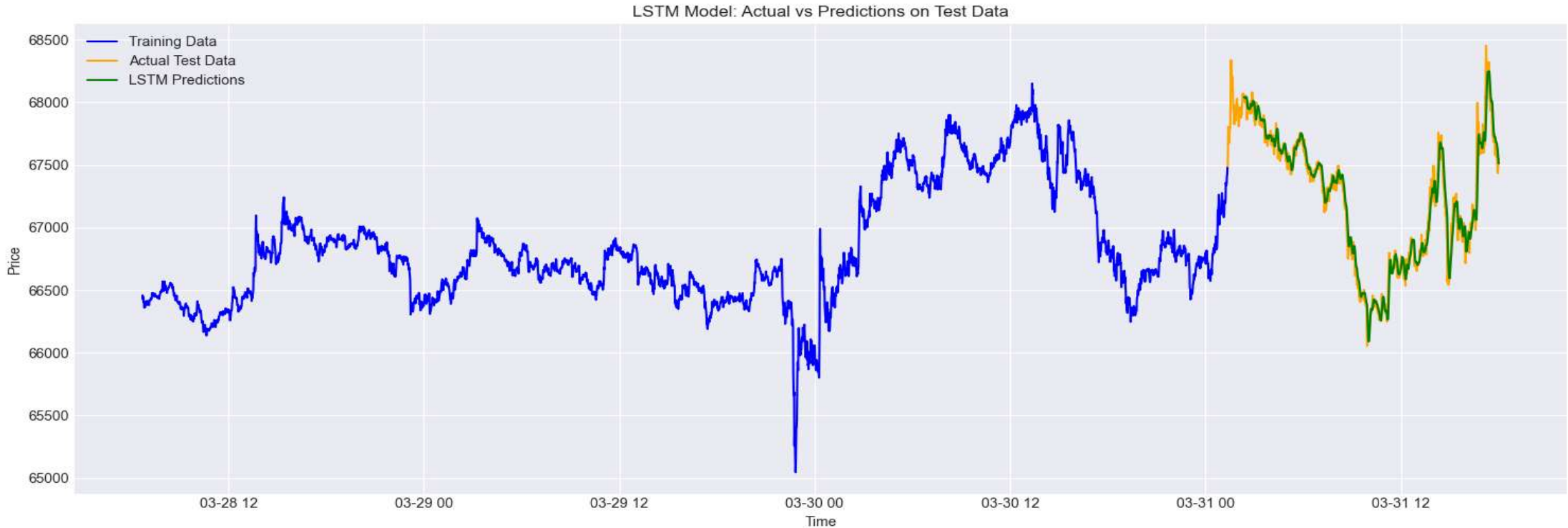


LSTM Model Performance

- The LSTM is evaluated using MAE, RMSE, and MAPE
- The values indicating predictions are very closer to actual values

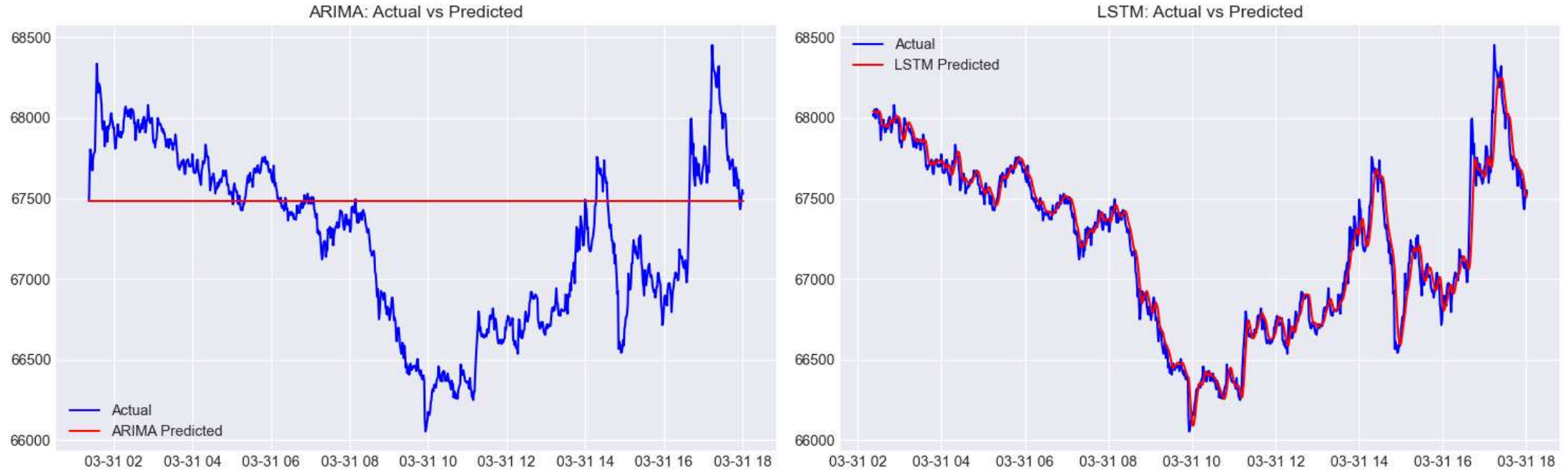
Model Performance	
MAE	64.41
RMSE	96.36
MAPE	0.09%

LSTM Actual vs Predicted graph



- The above graph shows LSTM model predictions closely follows the actual bitcoin prices
- The model is capturing well in both trend and short-term fluctuations
- This model is adapting to sudden changes

Model Performance (ARIMA vs LSTM)



- Arima produces both smooth and stable patterns and capturing overall trend in bitcoin prices but limited responses for short-term fluctuations.
- On other-side LSTM is closely capturing both trend and fluctuations the model is adapting to sudden changes

Conclusion

- In this project I have Implemented ARIMA and LSTM models
- ARIMA performed well in capturing overall Trends
- LSTM addressed limitations of ARIMA by capturing both trends and fluctuations
- The performance metrics showed LSTM have low error values
- ARIMA provides a strong baseline for time series analysis
- LSTM is more suitable for real-time cryptocurrency price forecasting since it has ability to handle complex and dynamic market behavior

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Thankyou